Solving Equations with Exponentials

Fluency · Problem Solving · Enrichment

*Big idea: an exponential equation can be solved without calculus or logarithms as soon as both sides are written as a power of the **same base**. Once that is true, the exponents themselves must be equal, turning the problem into an ordinary equation.*

1 Fluency

Multiplying exponentials ($a^m \times a^n = a^{m+n}$)

Simplify each expression.

1. $2^3 \times 2^4$
2. $x^5 \times x^2$
3. $5^2 \times 5^4 \times 5$
4. $4^2 \times 4^5$
5. $3 \times 3^4 \times 3^2$
6. $x^3 \times x \times x^4$

Dividing exponentials ($a^m \div a^n = a^{m-n}$)

Simplify each expression.

1. $\dfrac{5^7}{5^3}$
2. $\dfrac{x^9}{x^4}$
3. $\dfrac{3^8 \times 3^2}{3^5}$
4. $\dfrac{6^{10}}{6^6}$
5. $\dfrac{7^{12}}{7^5}$
6. $\dfrac{x^{11}}{x^3}$

The power law ($(a^m)^n = a^{mn}$)

Simplify each expression. For Questions 15 and 17, also evaluate your answer as a single number.

1. $(3^2)^4$
2. $(x^3)^5$
3. $(2^2)^3$
4. $(5^3)^2$
5. $(2^4)^3$
6. $(x^2)^6$

Writing exponentials with a common base

1. Write 32 as a power of 2.
2. Write 81 as a power of 3.
3. Write 64 as a power of 2.
4. Write $\frac{1}{9}$ as a power of 3.
5. Rewrite 9^x as a power of 3 (in the form $3^{\square}$).
6. Rewrite $\left(\frac{1}{3}\right)^x$ as a power of 3.

Combining related bases into a single base

Some expressions mix two different bases that are both powers of the same prime (for example, $8 = 2^3$). Rewrite every term with that prime as the base, then use the multiplying or dividing law to combine them into a **single power**.

1. Simplify $2^{3x+1} \times 8^{2x+8}$ as a single power of 2.
2. Simplify $9^x \times 3^{2x+5}$ as a single power of 3.
3. Simplify $4^{x+2} \times 2^{3x-1}$ as a single power of 2.
4. Simplify $\frac{16^{x+1}}{4^{x-2}}$ as a single power of 2.
5. Simplify $27^x \times 3^{x+2}$ as a single power of 3.
6. Simplify $\frac{8^{x+2}}{2^{x-1}}$ as a single power of 2.

Solving exponential equations

Once both sides are written as a power of the same base, equate the exponents and solve.

1. Solve $2^x = 32$.
2. Solve $5^x = \frac{1}{25}$.
3. Solve $3^{2x-1} = 27$.
4. Solve $4^x = 256$.
5. Solve $2^{x-1} = \frac{1}{8}$.
6. Solve $2^{3x+1} \times 8^{2x+8} = 2^{25}$. (Use your answer to Question 25.)

2 Problem Solving

Many exponential equations are not immediately ready to solve. First **rearrange** the equation so the exponential term is on its own, then rewrite both sides with the same base, then equate exponents and solve.

Rearranging to a common base

1. Solve $3^x + 5 = 14$.
2. Solve $2 \times 5^x - 3 = 47$.
3. Solve $\dfrac{4^x}{2} + 1 = 33$.
4. Solve $50 - 2^x = -14$.
5. Solve $9^x = 3^{x+4}$. (Both sides are already exponential — rewrite the left-hand side using base 3 before equating exponents.)
6. Solve $3^{x+3} \times 3^{2x} = 9^{x+4}$.

Leaving your answer in terms of k

In these problems, k is a fixed but unspecified number. Solve for x as usual, leaving your final answer as an expression in terms of k.

1. Solve $3^{2x-1} = 3^{k+5}$ for x in terms of k.
2. Solve $4^x = 8^k$ for x in terms of k. (Rewrite both sides using base 2 first.)
3. Solve $2^x \times 4^x = 2^k$ for x in terms of k.
4. Solve $\dfrac{8^{x+1}}{2^{x-3}} = 2^k$ for x in terms of k.
5. Solve $9^x = 3^{k+4}$ for x in terms of k.
6. Solve $\dfrac{2^{x+5}}{4^{x-1}} = 2^k$ for x in terms of k.

3 Enrichment

Factoring out a separate prime

When a base is a product of two primes (such as $6 = 2 \times 3$), dividing by a power of one of those primes can leave a power of the other. For example,

$$\frac{6^{x+3}}{3^{x+3}} = \left(\frac{6}{3}\right)^{x+3} = 2^{x+3}.$$

1. Use this idea to solve: $\dfrac{6^{x+3}}{3^{x+3}} = 2^{5x+1}$.
2. Solve: $\dfrac{10^{x+1}}{5^{x+1}} = 2^{3x-2}$.
3. Solve: $\dfrac{15^{x}}{5^{x}} = 3^{2x-4}$.
4. Solve: $\dfrac{14^{x-2}}{7^{x-2}} = 2^{3x-10}$.
5. Solve: $\dfrac{21^{x}}{7^{x}} = 3^{2x-8}$.
6. Solve: $\dfrac{33^{x-1}}{11^{x-1}} = 3^{2x+1}$.

All skills at once

1. Consider the equation
$$\frac{(4^x)^2}{2^3} \times \frac{20^{x+1}}{5^{x+1}} = 2^{2k-1}.$$
   (a) Simplify $\dfrac{(4^x)^2}{2^3}$ as a single power of 2.
   (b) Simplify $\dfrac{20^{x+1}}{5^{x+1}}$ as a single power of 2, by factoring out the prime 5.
   (c) Combine your two results and equate exponents to solve for x in terms of k.
   (d) Check your formula by substituting $k = 6$ into your expression for x, then substituting both values back into the original equation to confirm both sides are equal.

2. Consider the equation
$$\frac{20^{x+1}}{5^{x+1}} \times 2^{3x} = 2^{k+2}.$$
   (a) Factor out the prime 5 to simplify $\dfrac{20^{x+1}}{5^{x+1}}$ as a single power of 2.
   (b) Combine this with the 2^{3x} term, then equate exponents to solve for x in terms of k.
   (c) Check your formula by substituting $k = 10$ and confirming both sides of the original equation are equal.

Challenge

No scaffolding this time — decide for yourself how to simplify this equation before you try to solve it.

3. Solve for x: $\quad 2^{x+3} \times 3^{x+1} = 2^5 \times 3^{2x-1}$.